

Transient Electronics That Disappear



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In this era of electronics-driven lifestyle with well-established advantages of durable, long-lasting electronics, and when the use of electronic devices has invaded into the human body itself, thinking about transient electronics (that dissolve after some time without leaving a trace) and its applications is inevitable.

Millions of patients have benefited from the innovative development of electronic medical devices/gadgets such as pacemakers or medicine-dispensing agents implanted in the body for either diagnostic or therapeutic benefits. While these have revolutionised modern medicine, many a time these outlive their purpose in the human body and require surgical removal to avoid complication. However, rather than removing these devices through surgery, what if these could simply disappear?

This is the concept behind transient electronics, the newly-developed electronic devices that are designed to dissolve inside the body once these have served their purpose. Transient electronics offer robust performance, which is comparable to current devices, but these completely resorb

into their environment at a prescribed time, ranging from minutes to years, depending on the application.

The goal of the electronics industry has been to build durable devices that last long with stable performance, but many new opportunities open up once we start thinking about electronics that could disappear in a controlled and programmable way. It is the new way of looking at electronics. Scientists and engineers are

developing transient materials, which are special degradable polymer composite materials designed to quickly and completely melt away when a trigger is activated.

The technology could be useful for any application in which sensitive data is used. Some of these are military electronics equipment, credit cards and passports, and any application in which the electronic device is to be used for a defined period of time such as in bioelectronics, implantable electronics and environmental monitoring applications.

Development status

Previous research in the area has explored the use of transient materials to create dissolvable devices such as transistors, resistors and diodes. Researchers have also developed and tested transient resistors and capacitors.

The goal of research is to investigate how the rate of transiency could actually be controlled. They are experimenting with a blend of programmable biodegradable and transient insulating polymer films. They have found that by adding gelatin to the mix, dissolution can be slowed, while addition of sucrose speeds up the rate of transiency.

Using these special polymers, researchers were able to build and test an antenna that was capable of sending data and then completely dissolving itself when a trigger was activated. One constant in this experimentation with different composite structures is that the material maintains appropriate physical properties to function as a substrate for electronics.

The researchers are pioneers in the engineering of ultra-thin flexible electronic components. Only a few tens of nanometres thick, these tiny circuits, from transistors to interconnects, readily dissolve in a small amount of water or body fluid, and are harmlessly resorbed or assimilated.

Controlled degradation and transiency of materials is of significant importance in the design and fabrication of degradable and transient biomedical and electronic de-

A transient electronic device (Image courtesy: rational-trader.blogspot.in)

